

LESSON 9-4 • SESSION 1 • PRACTICE • FORM B

Substitute and Solve

Learning target – I can substitute a value into an equation to find the matching output. · 6.EE.C.9

REMEMBER

- **Substitute** means put a known number in place of a letter. Then do the math and add units.
- With a fee, **multiply first**, then add: $c = 3n + 10$ with $n = 4$ is $3 \times 4 + 10 = 22$.
- Check that the answer is **reasonable** and answers the question asked.

WORKED EXAMPLE

A bakery makes 24 rolls each hour: $r = 24h$. Is 5 hours enough for 150 rolls?

Step 1 Substitute $h = 5$: $r = 24 \times 5 = 120$ rolls.

Step 2 Compare: $120 < 150$.

Answer No, 5 hours makes only 120 rolls

WORDS substitute (*sustituir*) · equation (*ecuación*) · reasonable (*razonable*)

PRACTICE

show your work · label units

1. Substitute to find the output.

a. $y = 8x$; $x = 7$ _____

b. $c = 4h + 12$; $h = 5$ _____

c. $y = 6x + 5$; $x = 9$ _____

d. $y = 3x + 8$; $x = 20$ _____

2. **Soap Workshop.** A workshop makes 16 bars of soap every hour. A store orders 120 bars.

a. Write an equation for the bars y made in x hours. _____

b. Find the bars made in 11 hours. _____ bars

c. Is an 8-hour day enough for the order? Explain. _____

3. **Muffin Trays.** A bakery bakes 12 muffins on every tray. Write an equation for the muffins m on t trays. Find the muffins on 14 trays.

Hint Write the rule, then put in the number of trays.

Show your work.

Answer _____ muffins

4. **Steady Rower.** Keisha rows 40 meters every minute. Write an equation for the meters d in n minutes. Find the meters in 12 minutes.

Hint Replace the letter for minutes with the number.

Show your work.

Answer _____ meters

5. **Packing Crew.** A crew packs 9 boxes every minute. After 30 minutes, the crew says it packed 290 boxes. Does that claim fit the rate?

Hint Substitute the time, then compare with the claim.

Show your work.

Fits? _____

6. **Bike Shop.** A shop builds 1.5 bikes each hour. Write an equation for the bikes b in h hours. Find the bikes built in 16 hours.

Hint Multiply the decimal rate by the hours.

Show your work.

Answer _____ bikes

7. **With a Fee.** A rental costs $c = 12h + 30$ dollars for h hours. Find the cost for 5 hours.

Hint Multiply first, then add the fee.

Show your work.

Answer _____ dollars

8. **Print Shop.** A shop charges $c = 4n + 15$ dollars for n posters.

- a. Find the cost of 10 posters. _____ dollars
 b. Find the cost of 20 posters. _____ dollars
 c. Explain why doubling the posters does not double the cost.

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Mateo's Work.** A pump fills 8 jugs each minute. Mateo wrote $j = 8m$. Then he said 20 minutes would fill 28 jugs. Explain his mistake and find the correct number.

TEST PRACTICE

choose the best answer

10. For $y = 7x$, find y when $x = 9$.

A 16 **B** 63 **C** 79 **D** 97

11. To find c in $c = 7h + 4$ when $h = 3$, the first step is –

A add 7 and 4 **B** multiply 7 by 3 **C** add 4 and 3 **D** divide 7 by 3

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

95 T	60 Z	90 L	480 S	24 U	168 E	176 R	26 X	210 Q	52 J
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#2b	#3	#4	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____